Report on:

"Artificial Intelligence & Machine Learning – Task 2"

"Feature Engineering, Model Optimization & Performance Comparison"

Submitted by

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1. Introduction

Machine Learning models are rarely deployed successfully without proper preprocessing and performance evaluation. In real-world projects, data preparation, feature scaling, and model comparison play a crucial role in building accurate and reliable predictive systems.

This project focuses on building a **House Price Prediction system** using the California Housing dataset. The main objective is to apply feature engineering techniques, train multiple regression models, evaluate their performance using standard metrics, and select the best-performing model.

Unlike basic model training, this task emphasizes:

- Proper feature scaling
- Training multiple algorithms
- Objective model comparison
- Performance validation using evaluation metrics

This structured workflow reflects how professional Machine Learning engineers approach real-world problems.

2. Dataset Description

The project uses the **California Housing Dataset**, which contains real-world housing-related information collected from California districts.

2.1 Target Variable

- **Median House Value (HousePrice)**

This is the value we aim to predict using regression models.

2.2 Input Features

The dataset includes the following numerical features:

- Median Income (MedInc)
- House Age (HouseAge)
- Average Rooms (AveRooms)
- Average Bedrooms (AveBedrms)

- Population
- Average Occupancy (AveOccup)
- Latitude
- Longitude

Each row represents housing statistics for a particular district.

The dataset is well-suited for regression tasks because the target variable is continuous.

3. Methodology

This project follows a structured Machine Learning workflow consisting of the following steps:

3.1 Importing Required Libraries

The following Python libraries were used:

- pandas – data manipulation
- NumPy – numerical computations
- matplotlib – data visualization
- scikit-learn – model building and evaluation

These libraries support dataset handling, preprocessing, model training, and performance evaluation.

3.2 Data Loading

The dataset was loaded directly using scikit-learn's built-in function:

fetch_california_housing(as_frame=True)

The feature columns and target column were combined into a single DataFrame for easier processing.

3.3 Feature and Target Separation

Machine Learning models require input features (X) and output target (y) to be handled separately.

- X → All feature columns
- y → HousePrice column

This separation prepares the dataset for model training.

3.4 Feature Scaling (Critical Step)

The dataset contains features with different numerical ranges. For example:

- Population values can be in thousands
- Latitude values range around 30–40
- Median income has a completely different scale

If scaling is not applied, features with larger numeric ranges may dominate model learning.

To solve this issue, **StandardScaler** was used:

- Mean becomes 0
- Standard deviation becomes 1

This ensures fair learning across all features and improves model stability.

3.5 Train-Test Split

To evaluate model performance properly, the dataset was split into:

- 80% Training Data
- 20% Testing Data

The training data is used to train models, while the testing data is used to evaluate performance on unseen data.

This prevents overfitting and ensures realistic performance measurement.

3.6 Model Training

Three regression models were implemented:

1. Linear Regression

A basic regression model that assumes a linear relationship between features and target variable. It serves as a baseline model.

2. Ridge Regression

An extension of Linear Regression that includes L2 regularization. It helps reduce overfitting by penalizing large coefficients.

3. Decision Tree Regressor

A non-linear model that splits data into decision rules. It captures complex relationships between features and target variable.

Each model was trained using the training dataset.

4. Model Evaluation

Model performance was evaluated using two important regression metrics:

4.1 Root Mean Squared Error (RMSE)

RMSE measures the average magnitude of prediction error.

- Lower RMSE → Better model performance
- Sensitive to large errors

4.2 R^2 Score (Coefficient of Determination)

R^2 measures how well the model explains the variance in the target variable.

- Value ranges from 0 to 1
- Higher R^2 → Better explanatory power

5. Results and Performance Comparison

After training all models, their performance metrics were compared.

Model	RMSE	R^2 Score
Linear Regression	~0.745	~0.575
Ridge Regression	~0.745	~0.575
Decision Tree	~0.724	~0.599

5.1 Analysis of Results

- Linear Regression and Ridge Regression produced similar performance.
- Decision Tree achieved:
    - Lowest RMSE
    - Highest R^2 Score

This indicates that Decision Tree captured non-linear relationships in the dataset more effectively than linear models.

6. Visual Performance Validation

A scatter plot of Actual vs Predicted house prices was created for the best-performing model.

In the visualization:

- The red diagonal line represents perfect prediction.
- Points closer to the red line indicate better prediction accuracy.

The Decision Tree model showed reasonable alignment with the reference line, confirming its superior performance among the tested models.

7. Discussion

This project highlights several important Machine Learning principles:

1. Importance of Feature Scaling

Without scaling, model performance may become unstable.

2. Need for Model Comparison

Different algorithms perform differently on the same dataset.

3. Objective Evaluation

Performance metrics such as RMSE and R^2 provide measurable and reliable evaluation.

4. Real-World Workflow

The process followed in this project mirrors professional ML development:

- Data preparation
- Preprocessing
- Multiple model training
- Evaluation
- Model selection

8. Conclusion

This project successfully implemented a structured Machine Learning workflow for house price prediction using the California Housing dataset.

Three regression models were trained and evaluated. Based on performance comparison:

- The Decision Tree Regressor achieved the lowest RMSE and highest R^2 score.
- Therefore, it was selected as the best-performing model.

The project demonstrates how preprocessing techniques, especially feature scaling, and systematic model comparison significantly improve predictive performance.

This task provides practical experience in feature engineering, model optimization, and performance evaluation, which are essential skills in real-world Machine Learning applications.

9. Technologies Used

- Python
- Jupyter Notebook

- pandas
- NumPy
- scikit-learn
- matplotlib

10. Final Outcome

The project achieved the following:

- Applied feature scaling successfully
- Trained multiple regression models
- Compared performance objectively
- Selected best-performing model
- Validated performance visually

This completes the objectives of Artificial Intelligence & Machine Learning – Task 2.